

Project Design Phase

Solution Architecture – HouseRent Project

Date	15 February 2025
Team ID	
Project Name	
Maximum Marks	4 Marks

Solution Architecture

The **HouseRent** solution architecture is designed to provide a scalable, secure, and user-friendly platform for online property rental and booking management. It bridges the gap between tenants searching for rental properties and property owners who want to list and manage their properties efficiently. The architecture integrates modern web technologies to ensure seamless communication between users, the application, and the database.

The primary goal of the HouseRent solution architecture is to:

1. Solve Business Problems

- Simplify the property rental process.
- Eliminate the need for manual property searching and booking.
- Provide a centralized platform for tenants and property owners.
- Improve transparency and accessibility in rental transactions.

2. Define the Software Structure

The HouseRent application follows a **MERN Stack Architecture** consisting of:

Presentation Layer (Frontend)

- Developed using **React.js** and **Vite**.
- Provides responsive user interfaces for property browsing, booking, notifications, favorites, and owner dashboards.
- Handles user interactions and communicates with backend APIs.

Application Layer (Backend)

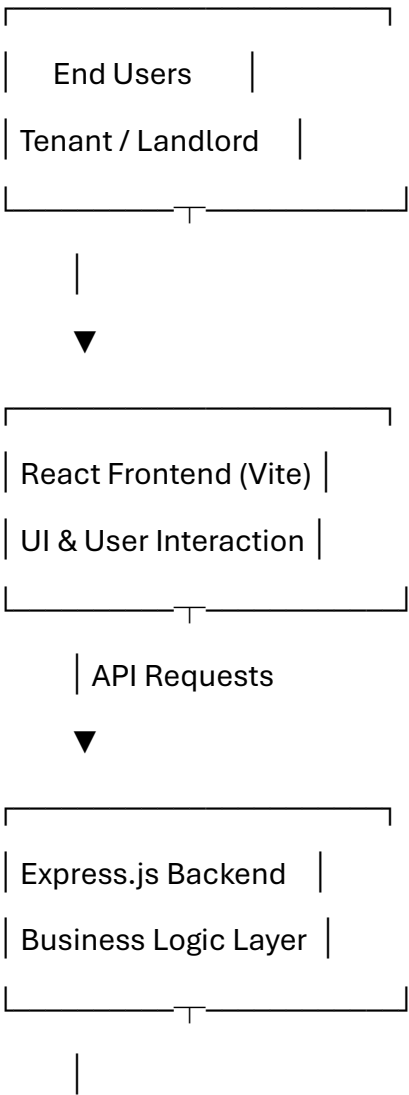
- Developed using **Node.js** and **Express.js**.

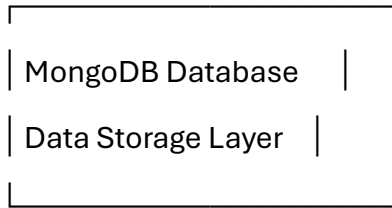
- Processes user requests and business logic.
- Manages property listings, bookings, favorites, notifications, testimonials, and blogs.
- Performs validation, security checks, and API routing.

Data Layer (Database)

- Implemented using **MongoDB** and **Mongoose**.
- Stores properties, bookings, notifications, favorites, testimonials, and blog information.
- Ensures reliable data persistence and retrieval.

Solution Architecture Diagram





Key Features Supported by the Architecture

Property Management

- Property listing creation.
- Property updates and deletion.
- Property browsing and filtering.

Booking Management

- Online booking creation.
- Booking history tracking.
- Booking status management.
- Booking cancellation.

Favorites Management

- Save favorite properties.
- Remove saved properties.
- Quick access to shortlisted properties.

Notification System

- Booking notifications.
- Payment notifications.
- Property-related alerts.
- Notification history tracking.

Owner Dashboard

- Add property listings.
- Manage properties.

- View rental inquiries.
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Development Phases

Phase 1: Requirement Analysis

- Identify tenant and landlord requirements.
- Define project objectives.

Phase 2: UI Design

- Design responsive user interfaces.
- Create user-friendly navigation and workflows.

Phase 3: Backend Development

- Develop REST APIs.
- Implement business logic and security.

Phase 4: Database Integration

- Design MongoDB collections.
- Implement data persistence.

Phase 5: Testing & Validation

- Functional testing.
- Performance testing.
- User acceptance testing.

Phase 6: Deployment

- Deploy frontend and backend services.
 - Configure production database.
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Solution Requirements

Functional Requirements

- Property search and filtering.
- Property booking.
- Booking history management.

- Favorites management.
- Notification management.
- Property owner dashboard.

Non-Functional Requirements

- Security
 - Scalability
 - Reliability
 - Performance
 - Responsiveness
 - Maintainability
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Benefits of the Proposed Architecture

- Modular and scalable design.
- Easy maintenance and future enhancements.
- Fast data retrieval and processing.
- Improved user experience.
- Secure communication between system components.
- Supports future integration of authentication, payment gateways, chat systems, and admin dashboards.

Conclusion

The HouseRent solution architecture provides a robust foundation for managing online rental operations. By combining React.js, Node.js, Express.js, and MongoDB, the platform delivers a secure, scalable, and efficient solution that addresses the needs of both tenants and property owners while supporting future growth and feature expansion.

Project Overview

Project Name

HouseRent – Online House Rental and Property Booking Platform

Project Description

HouseRent is a web-based rental property management platform developed using the MERN Stack (MongoDB, Express.js, React.js, and Node.js). The application enables users to search, view, compare, and book rental properties through an easy-to-use interface. It provides advanced filtering options, property details, favorites management, booking history tracking, notifications, and owner dashboard functionalities. The platform helps tenants find suitable rental properties while allowing property owners to efficiently manage their listings and rental activities.

Project Version

Version 1.0

Testing Period

Start Date: 01 March 2025

End Date: 15 March 2025

Testing Scope

Features and Functionalities to be Tested

- Home Page Navigation
- Property Search Functionality
- Property Filtering (Location, Price, Type, Amenities)
- Property Listing Management
- Property Details View
- Favorites Management
- Property Booking System
- Booking History Tracking
- Notification Management
- Owner Dashboard
- Add Property Feature
- Edit Property Feature
- Delete Property Feature
- Blog and Testimonial Modules
- Responsive User Interface

- API Integration
- Database Operations
- Error Handling and Validation

User Stories / Requirements to be Tested

Tenant/User Requirements

- Search rental properties.
- View property details and images.
- Save favorite properties.
- Book rental properties.
- Track booking status and history.
- Receive notifications and updates.

Property Owner Requirements

- Add new property listings.
- Edit existing property details.
- Delete property listings.
- Manage property information efficiently.

System Requirements

- Store and retrieve property data accurately.
- Maintain booking records.
- Provide fast and secure data access.
- Ensure responsive performance across devices.

Testing Environment

Hardware Requirements

- Processor: Intel Core i3 or above
- RAM: 4 GB or above
- Storage: 10 GB free space

Software Requirements

- Operating System: Windows 10 / Windows 11
- Browser: Google Chrome, Mozilla Firefox, Microsoft Edge
- Frontend: React.js, Vite
- Backend: Node.js, Express.js
- Database: MongoDB

URL / Application Location

Frontend URL:

http://localhost:5173

Backend API URL:

http://localhost:3000

Credentials (If Required)

Test User Account

Email: testuser@houserent.com

Password: Test@123

Property Owner Account

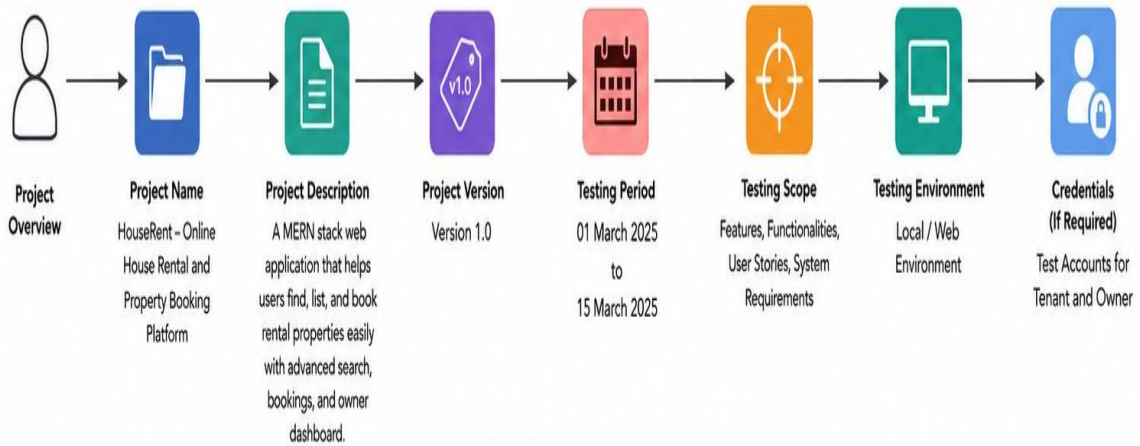
Email: owner@houserent.com

Password: Owner@123

Expected Outcome

The testing process will verify that all HouseRent functionalities operate correctly, including property discovery, booking management, favorites handling, notifications, and property administration. The system should provide a reliable, secure, responsive, and user-friendly experience for both tenants and property owners while maintaining data accuracy and system performance.

HouseRent Project Overview & Details



TESTING SCOPE



Features & Functionalities to be Tested

- ✓ Home Page Navigation
- ✓ Property Search
- ✓ Property Filtering
- ✓ Property Listing
- ✓ Property Details View
- ✓ Favorites Management
- ✓ Property Booking
- ✓ Booking History
- ✓ Notification Management
- ✓ Owner Dashboard
- ✓ Add / Edit / Delete Property
- ✓ Blog & Testimonial Modules
- ✓ Responsive UI
- ✓ API Integration
- ✓ Database Operations
- ✓ Error Handling & Validation
- ✓ Security & Authorization



User Stories / Requirements to be Tested

Tenant User Stories

- Search and view rental properties
- Save favorite properties
- Book properties online
- Track booking status and history
- Receive notifications and updates

Owner User Stories

- Add new property listings
- Edit existing properties
- Delete property listings
- Manage property information
- View booking and inquiries



System Requirements

- Store and retrieve data accurately
- Maintain booking records
- Ensure system performance
- Provide secure data access
- Ensure responsive experience

TESTING ENVIRONMENT



Application URL

http://localhost:5173
(Frontend)



API Base URL

http://localhost:3000
(Backend)



Database

MongoDB
(Local)



Tech Stack

MERN Stack
(MongoDB, Express.js,
React.js, Node.js)



Environment

Windows 10/11
Chrome / Edge /
Firefox

CREDENTIALS (IF REQUIRED)



Test Tenant Account

- Email: tenant@test.com
- Password: Tenant@123
- Role: Tenant / User



Test Owner Account

- Email: owner@test.com
- Password: Owner@123
- Role: Property Owner

Note: The above details are for testing purposes only. Please use valid test credentials and do not modify or delete test data.